SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: DEEP LEARNING COURSE CODE: MLA0408

COURSE OUTCOME COVERED

C02: Analyze Machine Learning and Deep Learning models by evaluating neural network architectures, representation learning, activation functions, unsupervised learning techniques, and their real-world applications. (L4)

CO Weightage:15%

Assessment Tool Weightage: 20%

QUESTIONS: Total Marks: 20

1. Differentiate between Machine Learning and Deep Learning with respect to architecture, data requirements, feature extraction and applications. Give two real-world applications of each.

2. Explain the concept of Representation Learning. Discuss how the width and depth of a neural network influence its learning capability, computational complexity and model performance.

3. Compare ReLU, Leaky ReLU (LReLU) and ELU (Exponential Linear Unit) activation functions. Explain their mathematical behavior, advantages, disadvantages and suitable use cases.

4. Explain the concept of unsupervised learning in neural networks. Describe the working principle of Restricted Boltzmann Machines (RBMs) and Autoencoders and compare their objectives and applications.

Code of Conduct:

I G. Naveen Kumar Reddy (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specifiedfor this assessment.
I understand that any violation ofacademic integrity rules will result in disciplinary action.

Signature of the student: G. Naveen Kumar Reddy

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 — Exemplary	Level 3 Proficient	Level 2 Developing	Level 1 — Beginning
Concept Understanding	25%	Clearly explains all concepts with complete accuracy and depth	Minor gaps in understanding	asic derstanding ith some errors	Incorrect or very limited understanding
Content Coverage	25%	Covers all equired topics comprehensively	Covers most topics with minor omissions	Covers some opics only	Very few topics covered
Technical Accuracy	20%	Il explanations are technically correct and recise	Minor technical errors	Some incorrect or ague explanations	Major technical errors
Integration of Concepts		Effectively connects and elates concepts 10 icall	Some connections made with minor a s	Limited connections etween conce ts	No clear relationship between conce ts
Clarity & Presentation		ell-structured, clear, and easy to nderstand	Mostly clear with minor issues	Somewhat nclear or oorly organized	Poorly structured and difficult to understand

CO-2 Assessment Tool-3

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Reg No:- 192525288.

Course Code :- MLA0408

Course Name :- Deep Learning.

Slot :- C

1.A Machine Learning and Deep Learning are two Important branches of Artificial Intelligence. used to Solve real world Problems. Machine learning use algorithms that Require Manually feature Extraction whereas Deep Learning Uses Multi-layer Neural Networks to automatically learn features from large amounts of data. Both Technologies are widely used in Solving real-world Problems Such as Image Recognition, Speech Processing, healthcare, finance, and Recommendation Systems. The following Table Compares ML and DL Based on Architecture, data Requirements, feature Extraction, and applications.

	Machine Learning	Le03tntn
Architecture	User Simple algorithms (île Decision Treer, SVM, RandOrf) Forest.	Uses multi-layer Artificial Neural Networks.
Data Requirement	10 with Small m datasets rneut•orn	Requires very datasets.
ceaJere	Manual feature engineering is needed	Automatically features from
Training Time	Faster	Slower due to many layers.
Hardware.	CPC),	Qe Requires GPUs for faster Training.
Ê)CCXL.$Q	Good for Structured Data	Better for Complex and Unstructured data

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Q.A Representation Learning is a technique in deep Learning where a neural network automatically Learns useful features from raw input data without manual features Engineering. It helps Improve prediction accuracy and Enables the model to identify Complex patterns in data.

Representation Learning.

* Automatically Extracts Meaningful features from raw data.
* Reduces the need for manual feature Engineering
* Improves the Performance of deep Learning Models.

Influence of Width and Depth.

Aspect	Width	Depth.
Learning Capability	Learns more features in a Single Layer	Learns hierarchical and Complex features.
Computational Complexity.	Increases memory Usage	Increases Training time and Computation.
Model Performance	Improves Performance up to an Optimal limit	Provides better accuracy for Complex Tasks but may Cause overfitting if too deep.

Advantages:

Extraction.

accuracy for Complex problems.

Manual Efforts in designing features.

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* Speech Recognition
* Natural Language Processing.
* Medical Image Analysis

Conclusion:-

Representation Learning Enables
to learn informative features utomatically. Th
width of a Network improves the ability to
learn more patterns, while the depth allows
to learn increasingly Complex Representations.

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Computational cost	$f(x)=max(0,x)$ put = 0 Output = x	001, S -A Small Negativ Output Output = x Lou)	$f(x)=x;\ x>0;;$ $\alpha(e^{x}-x)$ Small Negative Output Output = x lea

ReLU (Rectified Linear Unit)

Mathematical Behavior:

Advantages:

* Simple and fast
* Reduces Vanishing

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* Trains deep net

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Disadvantages:

* Suffers from the dying ReLU problem.
* Ignores Negative Input Values.

Suitable Use Cases:

* Deep CNNs
* Image classification.
* General deep Learning Tasks.

2. Leaky ReLU

Mathematical Behavior:

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Advantages

* Prevents the dying ReLU Problem.
* Allows a Small gradient for Negative
* Improves Learning Stability.

DisAdvantages:

* Requires Selecting the parameter α.
* Slightly more Computation than ReLU.

Suitable Use Cases:

dying ReLU problem Occurs, while ELU is Suitable for Models Requiring Stable Learning and higher accuracy despite increased Computation.

4.A Unsupervised Learning is a Machine Learning technique in which the Model Learns from Unlabeled data. Unlike Supervised Learning, there are no targets Outputs. The objective is to discover hidden patterns Structures in the data.

Working:-

* Input data is given without Labels.
* The Network identifies Similarities and patterns automatically.
* It Groups Similar data Representations.

Applications:-

* Clustering
* Dimensionality reduction.
* Feature Extraction
* Anomaly detection.
* Image Compression.

Restricted Boltzmann Machine.

An RBM is a generative Neural Network Used to Learn Probability distributions of input data. It Consists of two layers.

* Deep Neural Networks where ReLU neurons become inactive.
* GANs and Classification Models.

3. ELU (Exponential Linear Unit)

Mathematical Behavior:

* If $x > 0$, output $= x$.
* If $x \leq 0$, output $= \alpha(e^{x} - 1)$

Advantages:

* Produces Smooth Negative Outputs.
* Reduces bias shift.
* Faster Convergence and improved Learning.

Disadvantages:

* Computationally Expensive due to the Exponential function.
* Slower than ReLU.

Suitable Use Cases:

* Deep Learning Models Requiring faster Convergence.
* Networks where high accuracy and Stable Training are Important.

Conclusion:

ReLU is best for General deep Learning applications due to its Simplicity. Leaky ReLU is Preferred when the

* Visible Layer: Receives input data.
* Hidden Layer: Learns hidden features.

There are no Connections within the Same Layer Exist Only between Visible and hidden Layers.

Working Principle:-

1. Input data is Provided to the Visible Layer.
2. Hidden layer Extracts important features.
3. The hidden Layer reconstructs the input
4. The Network minimizes the reconstruction error by Adjusting weights.

Advantages:-

* Learns hidden features automatically.
* Useful for feature Extraction.
* Can Generate new data Samples.

Applications:-

* Recommendation Systems.
* Feature Learning.
* Dimensionality reduction.
* Deep Belief Networks.

Auto Encoder

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It has three parts:-

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* **Encoder:-** Compresses the input into a lower-di
representation.
* **Latent Space:-** Stores the Compressed features.
* **Decoder:-** Reconstructs the Original input from 44
Compressed Representation.

Working Principle:-

1. Input data is passed to the Encoder.
2. Encoder Creates a Compressed Representation.
3. Decoder Reconstructs the Original data.
4. Training Minimizes the Reconstruction Error.

Advantages

- Learns Efficient feature Representations.
* Removes Noise from data.
* Compresses high-dimensional data.

Applications:-

Image denoising.

Image Compression

Anomaly Detection

Feature Extraction.